MIL-C-82541B(OS)
21 April 1976
Superseding
MIL-C-82541A(OS)
23 August 1968

MILITARY SPECIFICATION

CRATE, UNDERWATER MINE MK 56 MOD 0
GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Sea Systems Command (OS), Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the requirements for manufacture and inspection of the Underwater Mine Crate, MK 56 Mod 0.

2. APPLICABLE DOCUMENTS

2.1 The following standards and drawings of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

STANDARDS

FED-STD-101	Preservation, Packaging, and Packing Materials, test procedures
MIL-STD-129	Marking for Shipment and Storage.

DRAWINGS, NAVAL SEA SYSTEMS COMMAND (CODE IDENT. 10001)

LD 550108	Crate, Underwater Mine, MK 56 Mod 0 List of Drawings, Assemblies, Parts, Specifications, etc.

(Copies of specifications, standards, and drawings required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 General Requirements

FSC 8140

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3.1.1 Construction. The Underwater Mine Crate, MK 56 Mod 0 shall be manufactured in accordance with LD 550108, including all documents listed thereon.

3.1.2 Precedence. In the event of conflict between applicable documents, the following order of precedence shall be observed:

a. Contract or Purchase Order
b. This Specification
c. Applicable Drawings of LD 550108
d. Referenced Documents of LD 550108
e. Other Applicable Documents Referenced in 2.1 of this Specification.

3.2 Interchangeability. All parts shall be capable of interchangeable assembly without interferences, binding or misalignment. Base Section, Top Section, Aft Support, Butt Plate assemblies and the lifting tubes at both ends of the base section should be inspected for interchangeability.

3.2.1 Instruments. The supplier shall provide necessary and adequate instruments to ensure that the crate will meet dimensions and requirements shown on the applicable documents listed in LD 550108.

3.3 First article. Before beginning quantity production under a contract, the supplier shall submit one first-article crate for complete inspection and test. This crate shall be manufactured in accordance with the procedure that is prepared for use in the execution of the contract. The first article will be inspected and tested by the Government as specified in 4.3. The inspection is for the purpose of determining that the supplier's production methods are capable of yielding items that comply with the technical requirements of the contract. The contracting officer will specify the location of the inspections and tests and will issue shipping instructions.

If the first article is considered satisfactory, the supplier will be so advised and the manufacture of crates intended for delivery under the contract may proceed. The supplier shall bear the expense of manufacture of this crate, which is not a part of the production quantity. This inspection and test shall not be construed as requalification or acceptance of subsequent production items.

3.4 Performance of first article. The first article will be subjected to all the inspections and tests prescribed in this section. Particular attention will be given to the material's structural strength, strength of welded joints, dimensional accuracy, interchangeability of parts, and general good workmanship. The Underwater Mine, MK 56, in its several configurations, will be placed in the crate without binding or misalignment of crate parts. All of the characteristics contained in applicable drawings of LD 550108 will be confirmed. Permanently distorted members and broken welds resulting from these tests which would prevent the reuse of the crate for protection, stacking, and handling of the Underwater Mine, MK 56, will constitute rejection of the first article.

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3.4.1 Vibration (repetitive shock). The crate in its normal shipping attitude with a dummy load as defined in 3.7 shall withstand the vibration (repetitive shock) test as specified in 4.6.1 and shall meet the acceptance criteria specified in 4.8.

3.4.2 Pushing and towing. The crate with a dummy load as defined in 3.7 shall be capable of pushing and towing. It shall be tested as specified in 4.6.2 and shall meet the acceptance criteria specified in 4.8.

3.4.3 Cornerwise-drop (rotational). The crate with a dummy load as defined in 3.7 shall withstand shock loading as induced by a rotational drop on stone, concrete, or steel covered concrete surface on each of the four base corners as specified in paragraph 4.6.3 and shall meet the acceptance criteria specified in 4.8.

3.4.4 Four-point hoisting. The four hoisting eyes shall be capable of withstanding a load of five times the gross weight of the crate and the dummy load as defined in 3.7. It shall be tested as specified in paragraph 4.6.4 and shall meet the acceptance criteria specified in 4.8.

3.4.5 Single-point hoisting. Each hoisting eye shall be capable of withstanding the gross weight of the crate and the dummy load as defined in 3.7. It shall be tested as specified in 4.6.5 and shall meet the acceptance criteria specified in 4.8.

3.4.6 Stacking. The crate with a dummy load as defined in 3.7 shall withstand a stacking load of five times the gross weight of the crate and dummy load defined in 3.7. It shall be tested as specified in 4.6.6 and meet the acceptance criteria specified in 4.8.

3.4.7 Impact. The crate with a dummy load as defined in 3.7 when released at a velocity of 10 feet per second, shall withstand impact on all four vertical faces against a rugged unyielding surface. It shall be tested as specified in 4.6.7 and shall meet the acceptance criteria specified in 4.8.

3.5 Performance characteristics of production units. Each crate made in production under the contract will be subjected to the quality conformance inspection of 4.5. All of the characteristics contained in Table I will be confirmed. A sampling of one crate from each lot shall be subjected to the test requirements prescribed herein. Particular attention will be given to the material's structural strength, strength of welded joints, dimensional accuracy, interchangeability of parts, and general good workmanship. Permanently distorted members and broken welds resulting from the test which would prevent the reuse of the crate for protection, stacking, and handling of the Underwater Mine, MK 56 will constitute rejection of production units.

3.5.1 Four-point hoisting. The four hoisting eyes shall be capable of withstanding a load of three times the gross weight of the crate and the dummy load defined in 3.7. It shall be tested as specified in 4.7.1 and meet the acceptance criteria specified in 4.8.

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3.6 Serial marking. The contracting agency or officer will assign the applicable serial numbers (in numerical sequence) to be applied to the identification label on each crate.

3.7 Dummy load. For test purposes, a dummy load (2026 pounds) simulating an Underwater Mine MK 56 Mod 0 (mine case, anchor, and explosive section) shall be used. Configuration and weight distribution shall be the same as a service or drill mine (mine case, anchor and explosive section), inert loaded.

3.8 Workmanship. Workmanship shall be equal to that required by best commercial practices consistent with quantity production of analogous parts, ensuring also that the dimensions, finishes, and tolerances specified herein and on the applicable drawings and specifications are satisfied. All components shall be sound and free from defects that would deleteriously affect the strength and performance of the crate, and all joints shall be tight and sound. All surfaces shall be free from rough spots, sharp edges and burrs, protrusions, dents or similar defects that might interfere with the assembly of the crate's components and loading of the crate, interfere with stacking, cause injury to personnel or damage to the contents during loading, handling and shipping. The workmanship on the first article will be the standard of quality which the production items must meet to be acceptable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classifed as follows:

4.2.1 First article inspection (see 4.3).

4.2.2 Periodic production inspection (see 4.4).

4.2.3 Quality conformance inspection (see 4.5).

4.3 First article inspection. The first-article unit that has satisfactorily passed the quality conformance inspection of 4.5 shall be subjected to all the tests of 4.6.

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4.4 Periodic production inspection. Units from lots that have satisfactorily passed the quality conformance inspections of 4.5 shall be selected for periodic production inspection. This inspection will be conducted at an activity to be designated by the procuring office. The manner of selection of the units shall be as follows:

Step 1. After production has begun, select at random one unit from the first lot of 50. The unit will be re-examined for compliance with Classification of Characteristics as specified on drawings, the workmanship requirement of 3.8 and the details applicable in LD 550108. If the unit satisfactorily passes this re-examination, proceed to Step 2.

Step 2. Same as Step 1 for each lot of 200 units.

4.4.1 Defective units. If a unit is determined to be defective as a result of the periodic production inspection, acceptance of the lot shall be withheld, and the contractor shall implement the following:

(a) Determine the cause of the defect.

(b) Take whatever actions are necessary to correct or eliminate the defect from all units in the lot.

(c) Submit to the procuring activity for its approval the proposed corrective action for preventing re-occurrence of the defect.

(d) If at Step 2, return to Step 1 again and conform with its requirements.

4.5 Quality conformance inspection.

4.5.1 Each crate shall be inspected to ensure fit of dummy load defined in 3.7.

4.5.2 Each crate shall be examined for compliance with details applicable in LD 550108 and Classification of Characteristics as specified on drawings.

4.5.3 Each crate shall be examined for workmanship in conformance with 3.8.

4.6 First-article tests. The crate, complete with dummy load installed as specified in 3.7, shall be subjected to the following tests at ambient temperature.

4.6.1 Vibration (repetitive shock) test. This test shall be conducted on the loaded crate in accordance with Method 5019 of FED-STD-101. It shall be tested with the base down in normal shipping position. The mine crate shall meet the requirements of 3.4.1. The test period shall be two hours. The following clarifications are applicable:

(a) If circular input motion is used, table frequency shall be adjusted to ensure that one edge of the container leaves the table on each cycle by 3/16 inch.

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(b) If vertical linear motion is used, all points of the container shall leave the table by 1/16 inch at the same instant during each cycle.

(c) If neither (a) nor (b) above is realizable with the equipment at hand, the table shall be operated at 270 ± 2 rpm.

4.6.2 Pushing and towing test. This test shall be conducted on a hard dry pavement in accordance with Test Method 5011 of FED-STD-101, paragraph 6.5 and 6.6. The mine crate shall meet the requirements of 3.4.2.

4.6.3 Cornerwise-drop test (rotational). This test shall be conducted on the loaded crate in accordance with Method 5005 of FED-STD-101. The drop heights shall be 18 inches on the four bottom corners of the crate only. The mine crate shall meet the requirements of 3.4.3.

4.6.4 Four-point hoisting test. A suitable equal four-legged sling with safety hooks shall be attached to the four lifting eyes on the bottom section of the loaded crate. A load of weights of 8800 pounds shall be placed on top of the crate with the dummy mine installed in the crate and the entire load shall be lifted free of the floor and suspended from the sling for five minutes. The mine crate shall meet the requirements of 3.4.4.

4.6.5 Single-point hoisting test. A suitable sling with a safety hook shall be attached to one of the lifting eyes on the bottom section of a crate and dummy mine. Each lifting eye shall be capable of lifting the total gross weight by single suspension free of the floor for five minutes. The mine crate shall meet the requirements of 3.4.5.

4.6.6 Stacking test. A load of 11,000 pounds shall be placed on top of the crate containing the dummy load. The load should be distributed and bear on the test crate so as to simulate loaded crates in a superimposed stack. The mine crate shall meet the requirements of 3.4.6.

4.6.7 Impact test. Pendulum impact test, Method 5012 of FED-STD-101, shall be used. The vertical drop shall be 18.6 inches. One impact shall be made on each end and side of the crate while suspended from a four-legged sling. The mine crate shall meet the requirements of 3.4.7.

4.7 Production unit test.

4.7.1 Four point hoisting test. A suitable equal four-legged sling with safety hooks shall be attached to the four lifting eyes on the bottom section of the loaded crate. A load of 4400 pounds shall be placed on top of the crate with the dummy mine installed in the crate and the entire load shall be lifted free of the floor and suspended by the sling for five minutes. The mine crate shall meet the requirements of 3.5.1.

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4.8 Acceptance criteria. Immediately after the tests of 4.6 are conducted on the first-article unit and tests of 4.7 on production units, those same units shall be examined for the following characteristics.

(a) There shall be no evidence of damage to the dummy load or shift of the contents within the crate that would create conditions likely to cause damage during subsequent shipment of the crated unit.

(b) There shall be no evidence of binding as the contents are removed from the crate.

(c) The crate shall not be bent or permanently distorted, welds broken or cracked or in any way damaged to the extent that the crate is not reusable for protection, stacking and handling of the underwater mine Mk 56 Mod 0.

(d) Any of the above characteristics is cause for rejection.

5. PACKAGING

5.1 Preservation packaging. Preservation and packaging shall be Level C. Unless otherwise specified, this crate is suitable for acceptance by common carriers without any special packaging. The electrical grounding clip shall be taped to a structural member.

5.1.1 Cleaning and preserving. All metal parts shall be wiped clean of flux, metal chips or other foreign matter. A light machine oil shall be applied to all screw threads.

5.1.2 Packing. The top and bottom sections shall be securely bolted together.

5.2 Marking. In addition to any special marking required by the contract, shipments shall be marked in accordance with MIL-STD-129.

NOTES AND CONCLUDING MATERIAL

6.1 Intended use. The Underwater Mine Crate Mk 56 Mod 0 is intended for use in packaging, handline, storage, and transportation of the Underwater Mine Mk 56 Mod 0 in several configurations as follows:

(a) Mine Case and Anchor

(b) Mine Case, Anchor, Explosive Section

(c) Mine Case, Anchor, Explosive Section and Flight Gear

(d) Mine Case, Anchor, Explosive Section, Flight Gear, and High Speed Fins and Nose Fairing.

In configurations (c) and (d) the crate functions as a storage cradle only.

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6.2 Description. The crate, of welded aluminum construction, primarily consists of rectangular tubes and channel sections, and is divided longitudinally in half so that it permits top opening. The upper weldment section secures to the lower half by means of swingbolt assemblies. The crate is provided with integral fork pockets capable of two-way forklift truck entry, and four lifting eyes to facilitate handling. The overall dimensions of the crate are 92 3/4" long, 27" wide, and 29 1/2" high. The weight of the empty crate is approximately 174 pounds.

6.3 Ordering Data. Procurement documents should specify the following:

(a) Title, number, and date of this specification

(b) Government activity to perform first-article inspections an tests.

(c) Quantity required

(d) Assignment of serial numbers.

Custodian
NAVY-OS

Preparing Activity:
NAVY-OS
Project No. 8140-N159

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DOCUMENT IDENTIFIER AND TITLE
MIL-C-82541B(OS) "Crate, Underwater Mine Mk56 Mod 0, Gen. Specification for

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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